4. Projectile Motion

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Turning a bang into a number

And working out a safety distance you can defend

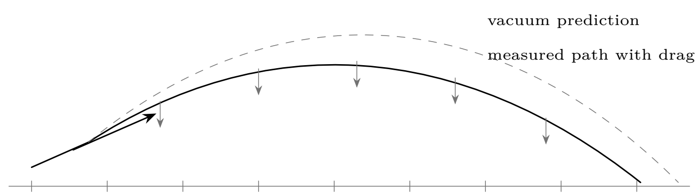
Lesson 4 · about 90 minutes · bench work, then range work with the adult

What you need

- A long tape measure or a measuring wheel
- A protractor or a phone inclinometer app
- A stopwatch, and a phone that shoots slow motion
- Marker flags or cones
- A notebook — this lesson is mostly recording

Predict first

At what launch angle does a projectile go furthest? Most people say 45 degrees. Write down whether you think that is exactly right for a potato, and if not, whether the best angle is higher or lower — and why.



Do it — on the bench first

1. **Drop and time.** Drop a ball from a measured height and time the fall. Repeat five times and average. Compute g from $h = \frac{1}{2}gt^2$. You should land near $9.8\,\mathrm{m/s^2}$.
2. **Horizontal launch.** Roll a ball off a table at various speeds. It always takes the *same* time to hit the floor. Horizontal and vertical motion are independent, and that single fact is the whole of projectile motion.
3. **Angle trial.** With a rubber-band launcher or a hose, fire at 20, 30, 40, 45, 50, 60 and 70 degrees and measure each range. Plot range against angle.

Then on the range

1. **Use a classroom projectile.** Roll a ball horizontally from a table into a clear floor area. Measure height and range; compute horizontal speed. Keep feet and faces out of the landing area.
2. **Or film the ball.** Put two scale marks in the same plane as the motion. Count frames and report frame-rate and parallax uncertainty.
3. **Angle series.** Use a low-energy classroom ball launcher or water stream within its instructions. Never use the combustion launcher to optimize angle or range.
4. **Compare with Lesson 3.** You predicted a muzzle velocity from the gas law. How close was it? If it is much lower, where did the energy go?

What it means

Two independent motions	Horizontal at constant speed, vertical accelerating downward. They do not affect one another, and every projectile problem is those two treated separately and then combined.
Range in a vacuum	$R = v^2 \sin(2\theta)/g$, which peaks at exactly 45 degrees. Notice the v^2: doubling muzzle velocity *quadruples* range. Small increases in fuel efficiency matter a great deal downrange.
Range in air	Not 45 degrees. Drag takes energy out for the whole flight, so a flatter, shorter flight loses less — the real optimum for a blunt, light projectile is usually somewhere between 35 and 42 degrees. Your measurements should show this, and it is a genuinely satisfying disagreement with the textbook.
A potato is a bad projectile	Light for its size, blunt, irregular and rough. Drag dominates, so it decelerates hard and falls well short of the vacuum prediction. Compare your measured range with v^2/g and see how much air took.
The energy	$\frac{1}{2}mv^2$. A 200 g potato at 100 m/s carries about 1000 J — comparable to a handgun round, with far more mass and less speed. This is the number that justifies every rule on the **Safety** sheet.
A trajectory is not a boundary	Never fire to discover maximum range and never double a measured range to invent a safety distance. Safety requires a walked empty range and positive earthen backstop independent of the equation.

Break it

Change the model projectile	Compare safe foam and rubber balls in a classroom apparatus. Record mass, diameter, and range. Never substitute projectiles in a combustion launcher.
Model versus data	Overlay low-energy trials on the vacuum prediction and graph the residual. A mismatch is a result, not a reason to add energy.
Predict, then check	Predict the next ball trial with an uncertainty interval, then measure it. The interval matters more than hitting a flag.
Wind	Fire the same shot with and against a steady wind. A light, blunt projectile is affected far more than they expect.

The point of the numbers

The reason to measure is to learn where a model succeeds and fails. A measured classroom velocity and a calculated trajectory become evidence with stated uncertainty, not permission to fire or a boundary for a real range. The model never replaces the walked range or positive earthen backstop.

Measurement and acceptance

Trial	h (m)	x (m)	frames	v_x (m/s)	uncertainty / landing observation
1					
2					
3					

Questions

Which measurements dominate uncertainty? Why does a side camera create parallax? Which assumptions fail for an irregular potato? Why can one accurate trial still fail to predict another?

ACCEPTANCE GATE

Pass only when learners distinguish a vacuum model, a low-energy trial, and a real range boundary. No combustion-launcher shot is authorized by this lesson.